6. DRY CONTACT



THERE IS ONE DRY CONTACT AVAILABLE ON THE CONNECTOR BOX. IT COULD BE USED TO RUN DC LOAD (5A) OR COULD BE USED TO DELIVER SIGNAL TO EXTERNAL DEVICE WHEN BATTERY VOLTAGE REACHES DC LOAD OFF LEVEL.

MPPT STATUS	CONDITION	DRY CONTACT PORT	
		NC & C	NO & C
POWER OFF	UNIT IS OFF & NO OUTPUT IS POWERED	CLOSE	OPEN
POWER ON	BATTERY VOLTAGE > DC LOAD OFF VOLTAGE	OPEN	CLOSE
	BATTERY VOLTAGE < DC LOAD OFF VOLTAGE	CLOSE	OPEN

7. USER SETTINGS MENU

PRESS AND HOLD THE ENTER BUTTON FOR ABOUT 3/4 SECONDS UNTIL SETTINGS MENU APPEAR ON THE LCD SCREEN.



PRESSING THE UP BUTTON WILL MAKE AN INCREMENT OF 0.1V IN THE RESPECTIVE VOLTAGE, SIMILARLY DOWN BUTTON WILL MAKE A DECREMENT OF 0.1V IN THE RESPECTIVE VOLTAGE.

AFTER MAKING THE DESIRED CHANGES, PRESSING THE ENTER BUTTON WILL MAKE A SHIFT TO THE NEXT SETTING.

1. BATTERY FULL VOLTAGE

   SELECTABLE BULK & CONSTANT VOLTAGE CHARGING RANGE (1st & 2nd STAGE CHARGING VOLTAGE)

12V MODE	24V MODE
UPTO 14.9V MAXIMUM	UPTO 29.9V MAXIMUM

2. FLOAT CHARGING VOLTAGE

   SELECTABLE FLOAT/3RD STAGE CHARGING VOLTAGE

HYBRID CHARGING MODE

WHEN EVER BATTERY CAPACITY DECREASES BELOW 50% OR SOLAR ENERGY IS INSUFFICIENT TO CHARGE THE BATTERY, MPPT SWITCHES TO **"HYBRID CHARGING MODE",** TURNS HYBRID MODE LED(BLUE) ON AND REMAINS IN **"HYBRID CHARGING MODE"** TILL BATTERY REACHES 100% CAPACITY.

BATTERY CHARGING MODES

THREE STAGE CHARGING TOPOLOGY IS IMPLEMENTED IN THE CONTROLLER TO INCREASE THE LIFE AND MAINTAIN THE GRAVITY OF BATTERY.

1. CC MODE OR CONSTANT CURRENT MODE.
   (BATTERY CHARGING LED(GREEN) KEEPS ON BLINKING AT AN INTERVAL OF 1 SECO
2. CV MODE OR CONSTANT VOLTAGE MODE.
   (BATTERY CHARGING LED(GREEN) KEEPS ON BLINKING AT AN INTERVAL OF 1 SECO
3. FLOAT MODE OR TRICKLE CHARGING MODE.
   (BATTERY CHARGING LED(GREEN) BECOMES STABLE AND A MESSAGE SHOWING **"BATTERY CHARGED"** APPEARS ON LCD SCREEN)

INSTALLATION

- THE WALL ON WHICH THE UNIT IS TO BE MOUNTED MUST BE STURDY AND CAN WITHSTAND THE WEIGHT OF THE UNIT.
- DO NOT INSTALL THE UNIT IN A BUILDING CONSTRUCTED OF FLAMMABLE OR HEAT RESISTANT MATERIALS.
- INSTALL THE UNIT ON AN EYE VIEW ORIENTATION TO FACILITATE INSPECTION OF THE LCD DISPLAY AND MAINTENANCE WORK.
- IT IS NOT RECOMMENDED TO EXPOSE THE UNIT DIRECTLY TO STRONG SUNLIGHT TO PREVENT OVERHEATING AND CAUSE POWER DERATING.

1. BATTERY CONNECTION

⚠ ALL WIRING MUST BE PERFORMED BY A PROFESSIONAL PERSON

⚠ CONNECTING THE BATTERY WITH SUITABLE CABLE's IS IMPORTANT FOR SAFE AND EFFICIENT OPERATION OF THE SYSTEM AND TO REDUCE THE RISK OF FIRE & INJURY.

⚠ BEFORE MAKING THE FINAL DC CONNECTION BE SURE POSITIVE (+) WIRE MUST BE CONNECTED TO THE POSITIVE (+) TERMINAL IN THE CONNECTOR BOX AND NEGATIVE (-) WIRE MUST BE CONNECTED TO THE NEGATIVE (-) TERMINAL IN THE CONNECTOR BOX. REVERSE POLARITY CONNECTION ON BATTERY WILL DAMAGE THE CONTROLLER.

1. PLEASE CHOOSE A SUITABLE BATTERY CABLE WITH CORRECT CONNECTOR WHICH CAN WELL FIT INTO THE BATTERY TERMINALS.
2. USE A SUITABLE SCREW DRIVER TO UNSCREW THE TERMINALS AND FIT THE BATTERY WIRES IN, THEN FASTEN THE TERMINALS BY SCREWDRIVER, MAKE SURE THE TERMINALS ARE WELL TIGHTENED.
   Note: A loose connection can damage the unit and increases the risk of fire.